

NextGen Exoskeleton Navigator Technical Report
NextGen Exoskeleton Navigator Security Requirements Exercise
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Executive Summary

The NextGen Exoskeleton Navigator (NGEN) system is a state-of-the-art exoskeleton system with capabilities of manned, unmanned, and autonomous modes with the means to support security, research, and engineering personnel. As the pinnacle of NextGen technology, the importance of its security cannot be understated. This system is designed to be deployed in detachments with the following:

- **Human Operators**
- **Maintenance Crews**
- **Drone Operations Center (DOC)**
- **Exoskeleton and Human Barracks**
- **Maintenance and Repair Shop**

These accompanying assets provide an invaluable support structure for the NGEN system. With this support structure, NGEN systems will be fully functional and thrive in hostile environments and remote geographic regions.

Security is of paramount concern in this NGEN project, with clearly defined cybersecurity countermeasures and their accompanying security requirements being identified prior to project development. Role-based-access control systems, data at rest and in transit encryption requirements, user authentication and traceability, and secure operational mode requirements being a few of many identified security requirements for the NGEN project.

In conclusion, the NextGen Exoskeleton Navigator System is the culmination of the greatest minds of our time to envision a product capable of propelling NextGen even further ahead of industry competition all while revolutionizing research, autonomous drone use, and security implementation. This report provides a detailed breakdown of some of the software requirements, security requirements, their relationships, and how they identify industry accepted security types to provide only the best foundations for a revolutionary system.

Discussion

Assumptions and Changes

Table 1: Assumptions and Changes

Assumptions and Changes ID	Description	Details
01	NGen Access Control Database Storage	Assumes that the NGen Access Control Database stores all roles designated to users.
02	NGen Analytics Software	The development of NGen Analytics Gathering software is suggested as an additional piece of the NGen design. This system will provide the remote gathering of NGen Exoskeleton analytics to allow for remote assessment of instruments, software, hardware, and features.
03	NGen Communications Software	The development of the Ngen Communications software is suggested as an additional critical component to this project. The systems purpose is to facilitate the communication between NGen Exoskeleton Navigators and DOC / Command Center personnel while operating out in the field.
04	NGen Local Data Storage	Assumes that NGen suits, while deployed into a field environment, will transmit data remotely to its Drone Operations

		Center attachment, while also maintaining a local database of data for redundancy purposes.
05	NextGen Exoskeleton Navigator Sensor Capabilities	Assumes the suit is equipped with capable and functional sensors and alert systems to detect and notify personnel whether it be observers, remote operators, or physical operators.
06	NextGen VPN	Assumes the NextGen organization will utilize or build a virtual private network to maintain the confidentiality of data travelling over connections.

Organizational Security Requirements

Table 2: Organizational Security Requirements

Security Requirement Policy	Description	AC
Internal Access Controls	Sensitive information of the classes internal, confidential, and restricted are to be restricted behind encryption and role-based access controls (RBAC). (IBM, 2021)	AC01
Data Integrity	The integrity of data within NextGen databases (research, engineering, security, access control, configuration) are to be maintained and properly stored in appropriate locations.	AC04
Acceptable Use Policy	Users functioning withing	

	NextGen systems and domains are to be held accountable for all their actions, of which will be recorded and stored within the Command Center security database.	
Network Security	The confidentiality and integrity of data in transit over NextGen network infrastructure is to be maintained.	
Remote Access	Connections to NextGen facilities, software, or systems from remote locations are to be executed via the NextGen VPN system.	AC06

Architectural Design Proposed Modifications

- **NextGen Exoskeleton Navigator (NGEN) Local Database:** NGEN suits are to be equipped with drives locally to store and maintain the data of sessions, research, security, pilot biometrics, and error logs. This is to provide redundancy in the event the transmissions of NGEN data to Drone Operations Centers are corrupted or interrupted.

Software Requirements

The following software requirements have been obtained from sources such as the NextGen Exoskeleton Inception Document, Organizational Security Policies, Architecture, Asset Inventory, and Regulatory Policy. Each derived software requirement has a requirement ID, source, name, description, and assumption and/or change. These functional or non-functional requirements serve as a means to build a solid foundation for the success of NGEN systems.

Table 3: Software Requirements

Req. ID	Source	Requirement Name	Description	AC
R001	DU	Pilot Role Adaptability	Adjust the systems accepted tool attachment use based on the given role of its pilot.	
R002	D	Human Operability Mode	Facilitate the operation of the NGEN Exoskeleton by a human pilot.	
R003	D	Remote Operability Mode	Facilitate the operation of the NGEN Exoskeleton via remote control from a Drone Operations Center (DOC)	
R004	D	Autonomous Operability Mode	Facilitate the operation of the NGEN Exoskeleton in an autonomous mode monitored from the Drone Operations Center (DOC)	
R005	D	Tool Flexibility and Interchange	Provide a method for pilots to attach and detach tools permitted by their role.	
R006	D	Sealed Mode	Support the locking and sealing of the NGEN Exoskeleton to facilitate sufficient conditions for pilot safety.	
R007	D	Roaming Operability	Provide a framework for NGEN Exoskeleton wireless connectivity to the DOC while operating in the field.	
R008	DR	Remote System Analysis and Monitoring	Facilitate the capture and wireless transmission of live system analytics to the DOC.	AC02
R009	DAI	Wired Connectivity	Provide the means for the NGEN Exoskeleton to connect via wired medium at a DOC exoskeleton barracks.	
R010	D	Wireless Pilot Communications	Facilitate the transmission and receiving of wireless communications from NGEN DOC and Command Centers.	AC03
R011	DS	Maintenance Operability Modes	Facilitate the transition from active to “repair” or “calibration” modes for maintenance personnel to maintain NGen Navigator tools.	
R012	DA	Windows Compatibility	Facilitate compatibility with NextGen Workstations which use the Microsoft Windows 11 operating system.	
R0013	DS	Operational Subnet Changeover	Support the transitioning and reacquisition of proper network information based on user	

			credentials.	
R0014	DRSA	Tool Attachment Requests	Provide users a method of submitting a tool attachment request to further increase the productivity or safety of NGen Navigator Systems.	
R0015	DSA	IAAA Verification	Ensure users attempting to access NextGen Navigator systems are permitted to do so prior to being given access to the Windows login screen for the software.	

Source – Requirements Source Codes:

D – Project Description | A – Architecture | I – Asset Inventory | U – Use Case |

R – Regulatory Policy | S – Organization Security Policy

Security Requirements

The following security requirements have been identified from sources such as the NextGen Exoskeleton Inception Document, Organizational Security Policies, Architecture, Asset Inventory, and Regulatory Policy. Each derived software requirement has a requirement ID, source, name, description, and assumption and/or change. These security requirements serve to build a solid foundation for the success of NGEN systems and apply to the industry accepted types of security requirements.

Table 4: Security Requirements

Security Requirement ID	Source	Security Requirement Name	Description	AC
SR001	DA	Role-Based Access Control	Permit authenticated users to only access features eligible for their designated role within NextGen systems.	AC01
SR002	DS	System Identification, Authentication and Authorization	Require at least two forms of authentication such as who they are, what they know, and where they are.	
SR003	DS	Suit Auditing, Traceability, and Accountability.	Ensure the tracing and logging of all actions by authenticated users while within NextGen Navigator systems for proper accountability.	
SR004	D	Tool Use Validation	Require user authentication prior to permitting a user to attach or use a tool. User's actions will	

			be recorded for auditing.	
SR005	DS	Secure Network Communications	All communications sent over wired or wireless mediums shall use secure and modern encryption systems to maintain the confidentiality of NextGen data.	
SR006	D	Safe Software Failure Countermeasures	All components of the NextGen software system shall fail safely to limit the exposure of sensitive data, prevent harm of human pilots, and protect company assets.	
SR007	D	Secure Maintenance Mode Operability	Facilitate the transition between active mode and maintenance mode, effectively locking all functionality out other than the absolute necessary to calibrate and maintain tools.	
SR008	DAS	Secure Network Segmentation	NextGen Exoskeleton systems shall be assigned a network address based on the designated role of their pilot, logically isolating different role groups	
SR009	DS	Deployment Session Archival	NextGen Exoskeleton systems, when deployed in the field, will record and archive all session data to the NGENius Access Control Database.	
SR010	DAS	Secure Local Data Storage	NextGen Navigator Systems will maintain a local copy of session, pilot, and action data as a redundancy should communication links fail.	AC04
SR011	DS	Secure Remote Access	The system will utilize secure communication techniques and session encryption, such as an organizational VPN, to ensure safe and secure operation of NGen Navigators.	AC06
SR012	D	System Failure Alert System	Provide remote operators and observers, or physical pilots notifications that a failure has occurred within the software system via visual or audible mediums.	AC05
SR013	DS	Input Validation	All forms of user input within NextGen software systems will validate the input against an accepted regular expressions and inputs.	

Source – Requirements Source Codes:

D – Project Description | A – Architecture | I – Asset Inventory | U – Use Case |

R – Regulatory Policy | S – Organization Security Policy

Types of Security Requirements to Security Requirements Matrix

The following table provides a visual relational matrix between NextGen Exoskeleton Navigator security requirements and industry accepted security requirement types.

Table 5: Security Requirements Matrix

Req.	C	I	A	T	AU	AO	SES	ER	CPM	ST	AR	INT	DE
SR001					AU	AO							
SR002	C				AU	AO							
SR003				T									
SR004					AU								
SR005	C	I											
SR006	C	I						ER					
SR007				T	AU		SES						
SR008	C	I				AO			CPM				DE
SR009							SES				AR		
SR010	C	I	A								AR		
SR011	C	I			AU	AO	SES						
SR012								ER					DE
SR013		I	A					ER					

C – Confidentiality | I – Integrity | A – Availability | T – Traceability | AU – Authentication |
AO - Authorization | SES – Session Management | ER – Error Exception and Management |
CPM – Configuration Parameters Management | ST – Sequencing and Timing |
AR – Archiving Requirements | INT – International | DE – Deployment Environment |

Software Requirements to Security Requirements Matrix

The following table provides a relational matrix between NGEN system software requirements and their respective security requirements. Multiple security requirements will apply to more than one software requirement serving as multiple layers of security throughout the NGEN system.

Table 6: Software and Security Requirements Matrix

SW Req.	SW Requirement Name	Security Req.	Security Req. Name
R001	Pilot Role Adaptability	SR001	Role-Based Access Control
R002	Human Operability Mode	SR002 SR003 SR006	System Identification, Authentication, and Authorization Suit Auditing, Traceability, and Accountability Safe Software Failure Countermeasures
R003	Remote Operability Mode	SR002 SR003 SR006	System Identification, Authentication, and Authorization Suit Auditing, Traceability, and Accountability Safe Software Failure Countermeasures
R004	Autonomous Operability Mode	SR002 SR003 SR006	System Identification, Authentication, and Authorization Suit Auditing, Traceability, and Accountability Safe Software Failure Countermeasures
R005	Tool Flexibility and Interchange	SR001 SR004 SR007	Role-Based Access Control Tool Use Validation Secure Maintenance Mode Operability
R006	Sealed Mode	SR006 SR012	Safe Software Failure Countermeasures System Failure Alert System
R007	Roaming Operability	SR005 SR008 SR011	Secure Network Communications Secure Network Segmentation Secure Remote Access
R008	Remote System Analysis and Monitoring	SR005 SR010 SR012	Secure Network Communications Secure Local Data Storage System Failure Alert System
R009	Wired Connectivity	SR005	Secure Network Communications
R010	Wireless Pilot Communications	SR005	Secure Network Communications

R011	Maintenance Operability Modes	SR007	Secure Maintenance Mode Operability
R012	Windows Compatibility	SR001	Role-Based Access Control
R013	Operational Subnet Changeover	SR001 SR002 SR009	Role-Based Access Control System Identification, Authentication and Authorization Deployment Session Archival
R014	Tool Attachment Requests	SR001 SR004	Role-Based Access Control Tool Use Validation
R015	IAAA Verification	SR001 SR002	Role-Based Access Control System Identification, Authentication and Authorization

Conclusions and Recommendations

The NextGen Exoskeleton Navigator (NGEN) system is a revolutionary technology that will change the way we think about robotics, autonomous systems, and biorobotics. With deployment environments of the best conditions to the worst conditions, the foundations to inception of this system are critical for its success. These software and security requirements have been identified and selected because they serve vital purposes in the overall success of this project. Each individual requirement, security or software based, builds a stable foundation for future expansions on the NextGen Exoskeleton Navigator system and their critically must be recognized. This report has provided a detailed deconstruction of software requirements whether they be functional or non-functional, security requirements and security requirement types, and their overall relationships to one another.

Recommendations:

- **Continuous Security Analysis:** The concept of security is not something to be done once and forgotten about. Continuous analysis of hardware, software, and their threats and vulnerabilities should be undertaken throughout the development process from start to finish, as well as after their production launch.
- **Incident Response Planning:** Incident response is a critical functionality of business systems. With this, incident response plans should be routinely updated and developed as needed (Adsero Security).
- **Disaster Recovery Planning:** Disasters are always a possibility, and the risk rating of threats will always be non-zero. From this, disaster recovery plans should be developed and consistently revisited to ensure the success and recovery of the NGEN project should disaster strike (Adsero Security).

References

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